NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

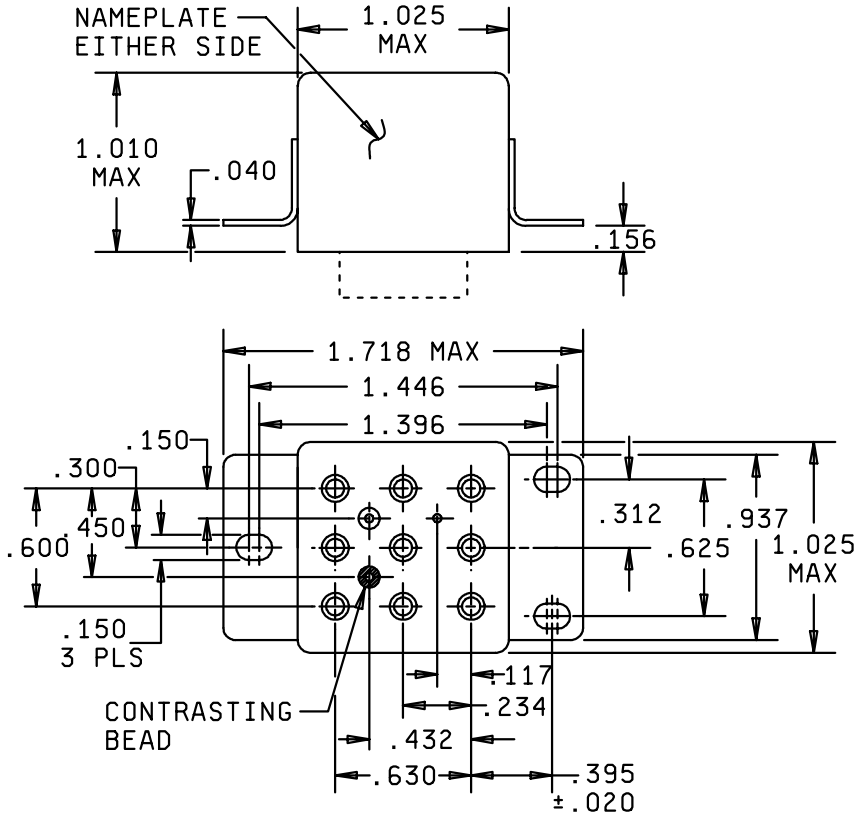
MIL-PRF-6106/13D(USAF)
26 November 1986
SUPERSEDING
MIL-R-6106/13C(USAF)
28 December 1984

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, 25 AMPERE, 3PST, NO, WITH 2 AMPERE 1PDT AUXILIARY CONTACTS, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE, TYPE I

This specification is approved for use within the Electronic Support Division AFLC, Departments of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-PRF-6106.



RAISED VERTICAL FLANGE MOUNT I AND TERMINAL LAYOUT
FOR -001, -002, -003, -005, -007, -008, -016 AND -017

FIGURE 1. Relay, outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

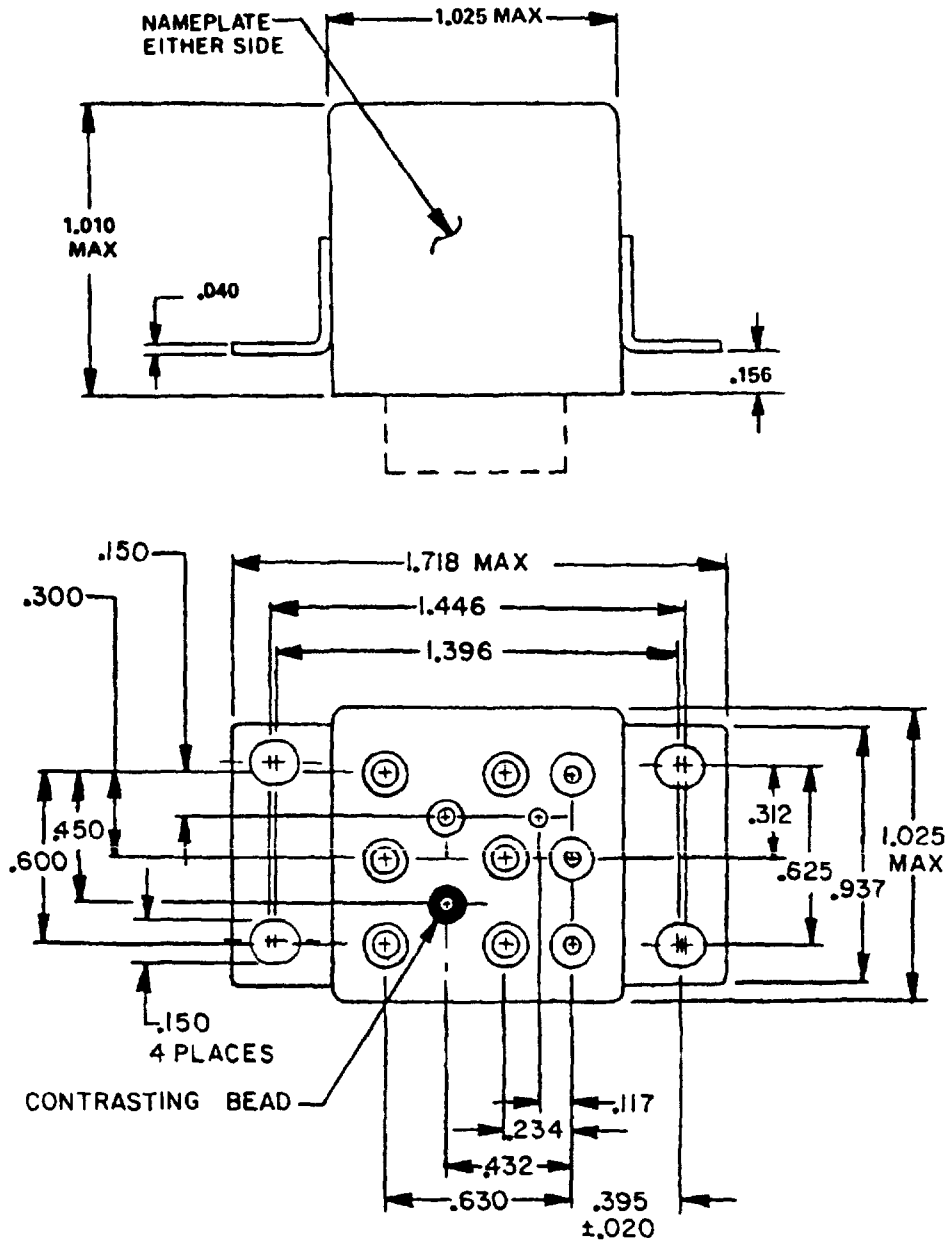
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm
.000	0.00	.150	3.81	.625	15.88
001	0.03	156	3.96	.630	16.00
.002	0.05	.234	5.94	.937	23.80
.003	0.08	.300	7.62	1.010	25.65
.020	0.51	.312	7.92	1.025	26.04
.040	1.02	.395	10.03	1.396	35.46
.062	1.57	.432	10.97	1.446	36.73
.093	2.36	.450	11.43	1 718	43.64
.117	2.97	.600	15.24		

NOTES:
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3 Unless otherwise specified, tolerance is ± .010 (.25 mm)

FIGURE 1 Relay, outline drawing- continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.



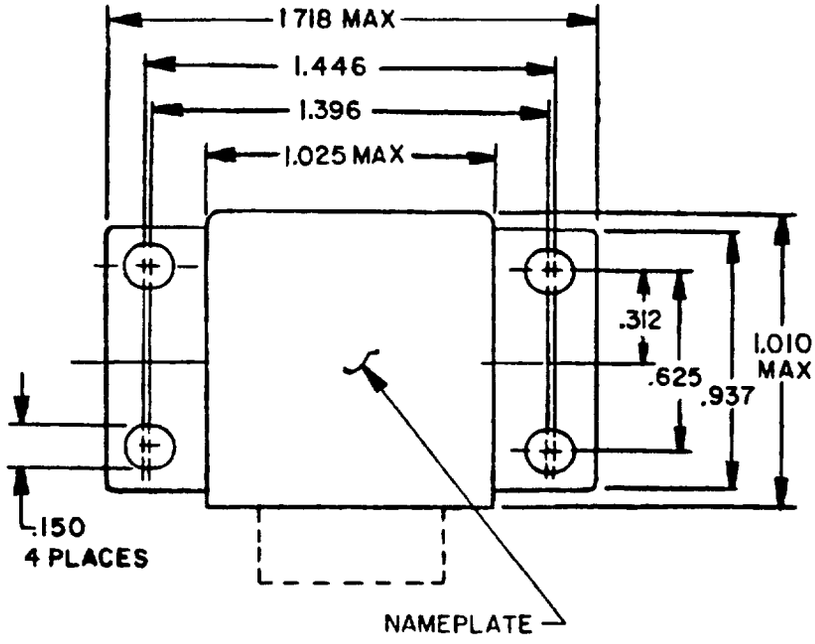
RAISED VERTICAL FLANGE MOUNT 2 AND TERMINAL LAYOUT FOR -004

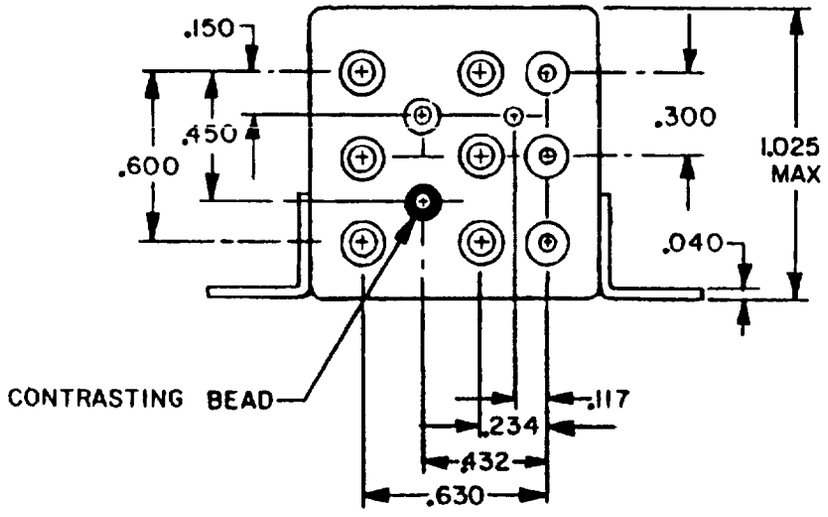
Inches	mm	Inches	mm	Inches	mm
.000	0.00	.150	3 81	.600	15.24
.001	0 03	.156	3.96	.625	15.88
.002	0.05	.234	5.94	.630	16.00
.020	0 51	300	7.52	937	23.80
.040	1 02	.312	7.92	1 010	25.65
062	1.57	.395	10.03	1.025	26 04
.093	2.36	.432	10.97	1.396	35.46
.117	2.97	.450	11.43	1.446	36.73
				1.718	43 64

NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± .010 (25 mm)

FIGURE 2 Relay, outline drawing

Source: https://assist.dla.mil -- Downloaded: 202610-17T06:23Z
Check the source to verify that this is the current version before use.





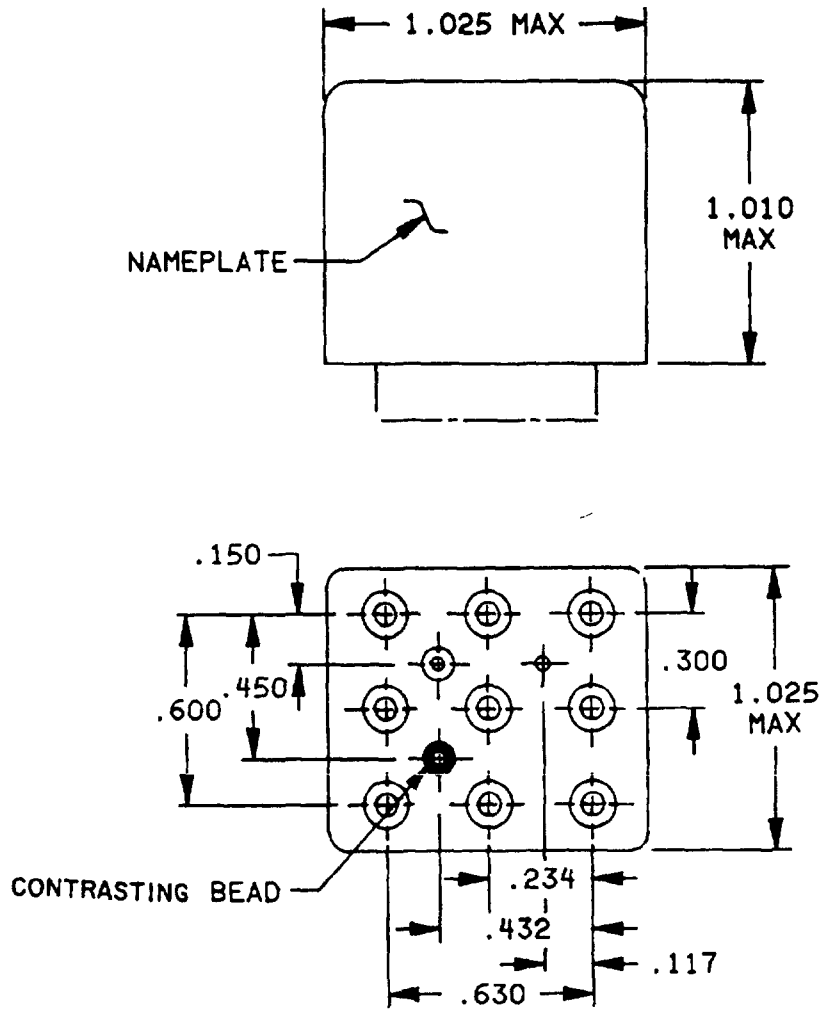
HORIZONTAL FLANGE MOUNT 3 AND TERMINAL LAYOUT FOR -006, -009 AND -018

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.150	3.81	.625	15.88
.002	0.05	.234	5.94	.630	16.00
.003	0.08	.300	7.62	.937	23.80
.040	1.02	.312	7.92	1.010	25.65
.062	1.57	.432	10.97	1.025	26.04
.093	2 36	.450	11.43	1.396	35.46
.117	2.97	.600	15.24	1.446	36.73
				1.718	43.64

NOTES.
1. Dimensions are in inches.
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± 010 (25 mm).

FIGURE 3 Relay, outline drawing.

Source: https://assist.dla.mil -- Downloaded: 202640-17T06:23Z
Check the source to verify that this is the current version before use.



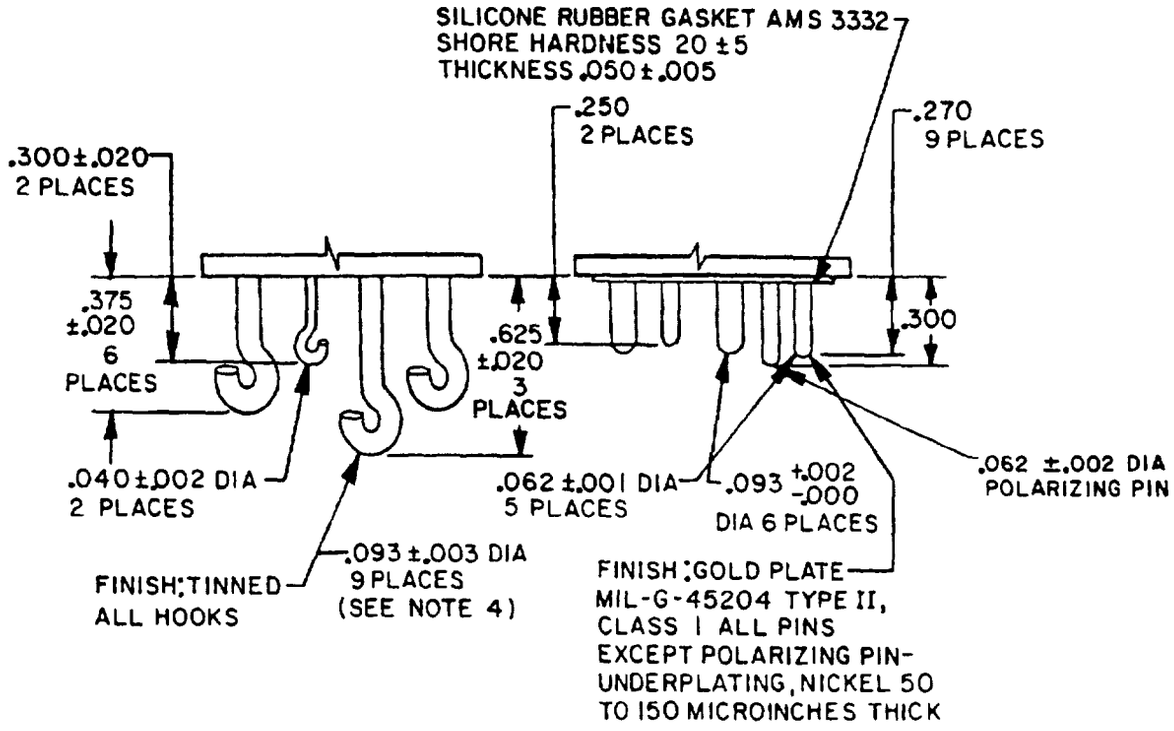
NO MOUNT 4 AND TERMINAL LAYOUT FOR -010, -011, -012, -013, -014, AND -015

Inches	mm	Inches	mm
.001	0 03	.234	5.94
.002	0.05	.300	7.62
.003	0.08	.432	10.97
.062	1 57	450	11.43
.093	2.36	.600	15.24
.117	2.97	.630	16 00
.150	3 81	1.010	25 65
		1 025	26 04

NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± 010 (25 mm)

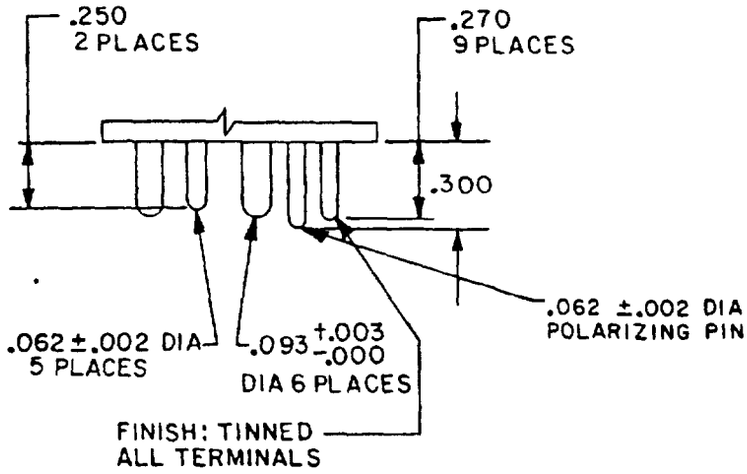
FIGURE 4 Relay, outline drawing

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.



SOLDER-HOOK TERMINAL

SOCKET/PLUG-IN TERMINAL



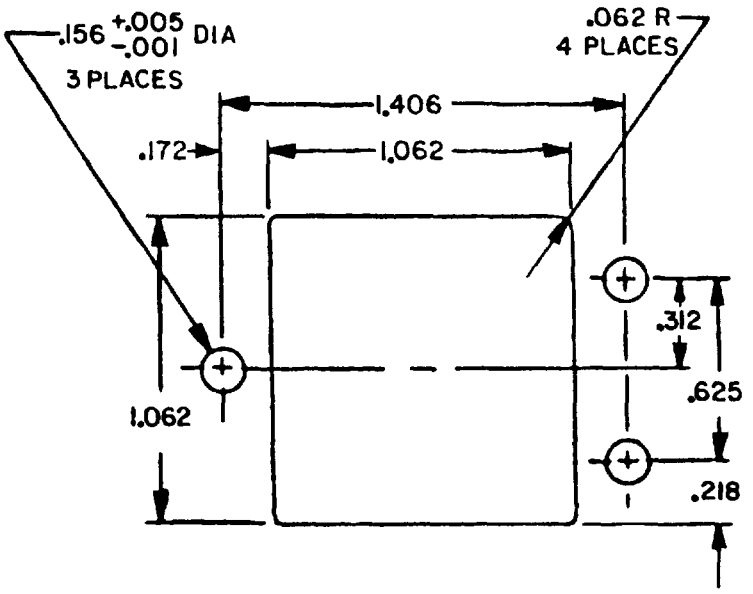
(D) SOLDER PIN TERMINALS

Inches	mm	Inches	mm	Inches	mm
000	0.00	020	0 51	.250	6.35
001	0.03	.040	1 02	270	6 86
002	0 05	050	1 27	.300	7.62
003	0 08	.062	1 57	375	9 52
005	0 13	093	2 36	625	15.88

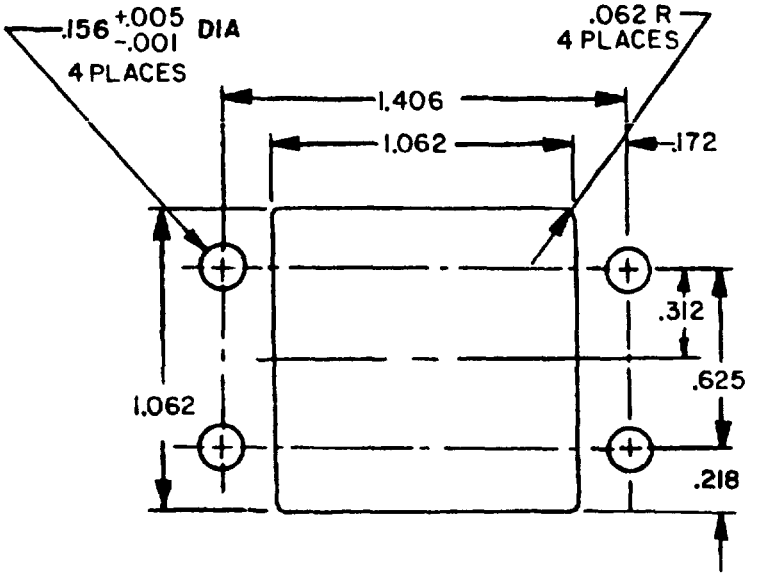
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Hook terminals will accept (1) one number 12 AWG stranded wire.

FIGURE 5 Relay, termination styles

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.



MOUNTING HOLE LAYOUT FOR FIGURE 1



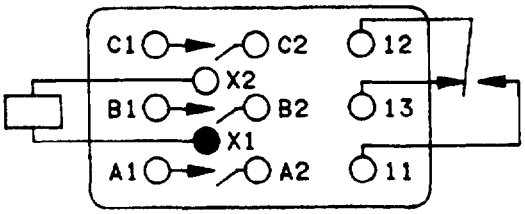
MOUNTING HOLE LAYOUT FOR FIGURES 2 AND 3

Inches	mm	Inches	mm
.001	0.03	.218	5.54
.005	0.13	.312	7.92
.062	1.57	.625	15.88
.156	3 96	1.062	26.97
.172	4.37	1.406	35.71

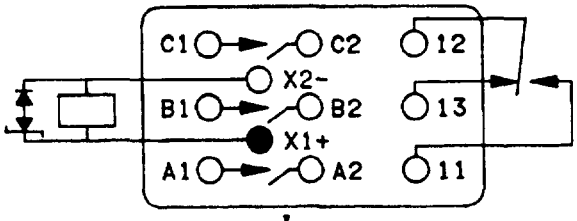
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .010 (.25 mm).

FIGURE 6 Relay, mounting hole layouts

Source: https://assist.dla.mil -- Downloaded: 202640-17T06:23Z
Check the source to verify that this is the current version before use.

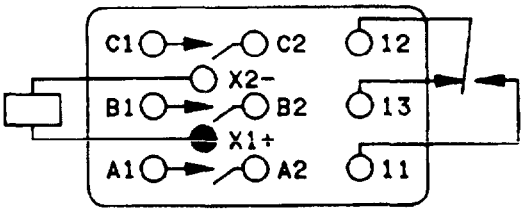


CIRCUIT DIAGRAM
MONOSTABLE
FOR -003, -004,
-012, AND -013
(SEE NOTE 3)



CIRCUIT DIAGRAM
POLARIZED MONOSTABLE
FOR -007, -008,
-014, -015, -016, -017 AND -018
(SEE NOTES 2 AND 3)

(D)



CIRCUIT DIAGRAM
POLARIZED MONOSTABLE
FOR -001, -002,
-005, -006, -009,
-010, AND -011
(SEE NOTE 2)

NOTES:
1. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
2. DC versions of this relay must not operate or be damaged by reverse polarity, semiconductors shall not be used for this purpose.
3. Coil polarity not applicable to AC versions -003, -004, -012, and -013.
4. When semiconductors are required, JANTX or equivalent screened semiconductors shall be used. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.

FIGURE 7 Relay, circuit diagrams

Source: https://assist.dla.mil -- Downloaded: 202640-17T06:23Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Coil data:

Coil data: See table I.

Duty rating: Continuous.

Operational data:

Rated contact load: See table II.

Operate time: See table I.

Release time: See table I.

Physical:

Dimensions and configuration: See figures 1 through 6.

Weight (maximum): 0.18 pound, 81.64 grams.

Terminal type: See figure 5 and table I.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C 1/.

Maximum altitude: 300,000 feet.

Shock G-level: 200 G, 100 G for -006 and -009 only.

Duration: 6 ms.

Maximum duration contact opening. 10 µs.

Vibration - sinusoidal:

G-level: 30 G, 20 G for -006 and -009 only.

Frequency range. 10 - 3,000 Hz.

Vibration - random.

Applicable standard: MIL-STD-202, method 214.

Test condition: IG, IE for -006 and -009 only.

Power spectral density: 0.4 G^2/Hz, 50-2,000 Hz (0.2 G^2/Hz for -006 and -009 only).

Duration. 15 minutes, each plane.

Acceleration: 15 G.

1/ For full rated load, maximum temperature, and altitude use no. 12 wire or larger. Solder hook relays are to be mounted to limit mounting bracket temperature to 160°C for dc units and 135°C for ac units.

Source: https://assist.dla.mil -- Downloaded: 202610-17T06:23Z
Check the source to verify that this is the current version before use.

ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests. 50 megohms.

Dielectric withstanding voltage (sea level): 2 to 5 seconds.

	Initial	After life tests
Coil to case	1000 V rms	1000 V rms
Auxiliary contacts	1000 V rms	750 V rms
All other points	1250 V rms	1000 V rms

Dielectric withstanding voltage (altitude) 2/: 1 minute.

	80,000 ft	300,000 ft
Coil to case	350 V rms	500 V rms
Auxiliary contacts	350 V rms	500 V rms
All other points	350 V rms	500 V rms

Maximum contact drop, initial. 0.150 volt.

After life test 0.175 volt.

Overload current: 50 amperes dc, 80 amperes ac.

Rupture current: 60 amperes dc, 100 amperes ac.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461.

Part number: M6106/13-(dash number from table I).

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.

Ⓓ TABLE I. Mounting, termination, and operating characteristics.

Dash no.	Coil terminal	Coil data: Nominal Volts	Nominal Freq Hz	Nominal Res	Maximum Volts	Maximum Amp	Max pickup voltage: Normal 1/	Max pickup voltage: High temp test	Max pickup voltage: Cont current test	Drop out voltage	Hold voltage	Time (milliseconds) (max): Operate 2/	Release 3/	Bounce: Main NO	Bounce: Main NC	Bounce: Aux NO	Bounce: Aux NC	Terminals	Mounting figure
-001																		Solder-hook	1
-002																		Socket plug-in	1
-005 4/																		Solder pin	1
-006 4/	DC	28	DC	-	29	.12	18	19.8	22.5	1.5	7.0	15	15	1.0	-	4	4	Solder pin	3
-009																		Solder-hook	3
-010																		Solder pin	4
-011																		Socket plug-in	4
-003																		Solder-hook	1
-004	AC	115	400	-	122	.04	90	95	103	5	30	20	50	1.0	-	4	4	Socket plug-in	2
-012																		Solder pin	4
-013																		Socket-plug-in	4
-007 5/																		Solder-hook	1
-008 5/	DC	28	DC	-	29	.12	18	19.8	22.5	1.5	7.0	15	15	1.0	-	4	4	Socket plug-in	1
-014 5/																		Solder pin	4
-015 5/																		Socket plug-in	4
-016 5/																		Solder pin	1
-017 4/5/																		Solder pin	1
-018 5/																		Solder hook	3

1/ Over temperature range.
2/ With nominal coil voltage.
3/ From nominal coil voltage.
4/ For -005, -006, and -017, quality conformance inspection shall be 100 percent for vibration scan, operational reliability, and contact bounce, operating and release times.
5/ Coil suppression (transient voltage back EMF) 42 V dc maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-01-11T06:23Z
Check the source to verify that this is the current version before use.

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive 2/	50	25		2	2	25		2		25			
Inductive	10	12		1	1								
Inductive	20					15		1		15			
Motor	50	10				10				10			
Lamp	50	5		.5	.5	5		.5		5			
Transfer load 3/													
Mechanical life Reduced current	200	6				6				6			
Min current	Applicable per specification (main contacts only)												

1/ Absence of value indicates relay is not rated for 3 phase application.
2/ For full rated load, maximum temperature, and altitude use no. 12 wire or larger. Solder hook relays to be mounted to limit mounting bracket temperature to 160°C for dc units and 135°C for ac units.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.

Ⓓ TABLE III. Qualification by similarity.

Part number M6106/13-	Loads		Dynamics 1/				Environmental	
	2/ A	2/ B	2/ A	2/ B	2/ C	2/ D	3/ A	3/ B
001	4		2				4	
002	4		2				4	
003		2, 3	1					2, 3
004		2, 3		1				2, 3
005	4		2				4	
006	4				2		4	
007	2, 3		1				2, 3	
008	2, 3		1				2, 3	
009	4				2		4	
010	4					2	4	
011	4					2	4	
012		2, 3				1	3	
013		2, 3				1	2, 3	
014	2, 3					1	2, 3	
015	2, 3					1	2, 3	
016	2, 3		1				2, 3	
017	2, 3		1				2, 3	
018	4				2		4	

1/ All units must be tested: See "dynamics" section of MIL-R-6106, Appendix.
2/ Terminals - socket pin type terminals are considered worse case. Solder pins and hooks may be qualified by similarity.
3/ Socket pins (with gasket seals) are considered different from solder pins and hooks, and each must be tested separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F631)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:23Z
Check the source to verify that this is the current version before use.